

### Q1.

A small, hollow, plastic ball of mass  $m$  kg is at rest on a point  $O$  on a polished horizontal surface. The ball is attached to two identical springs. The other ends of the springs are attached to the points  $P$  and  $Q$  which are 1.8 metres apart on a straight line through  $O$ .

The ball is struck so that it moves away from  $O$ , towards  $P$  with a speed of  $0.75 \text{ m s}^{-1}$ .

As the ball moves, its displacement from  $O$  is  $x$  metres at time  $t$  seconds after the motion starts.

The force that each of the springs applies to the ball is  $12.5mx$  newtons towards  $O$ .

The ball is to be modelled as a particle. The surface is assumed to be smooth and it is assumed that the forces applied to the ball by the springs are the only horizontal forces acting on the ball.

- (a) Find the minimum distance of the ball from  $P$ , in the subsequent motion.

(5)

- (b) In practice the minimum distance predicted by the model is incorrect.

Is the minimum distance predicted by the model likely to be too big or too small?

Explain your answer with reference to the model.

(2)

(Total 7 marks)

### Q2.

A railway truck, of mass  $m$ , is travelling in a straight line along a horizontal track. At time  $t = 0$ , the truck strikes one end of a buffer which is fixed at its other end. The buffer may be modelled as a light spring of natural length  $a$  and modulus of elasticity  $am n^2$ , where  $n$  is a positive constant. At time  $t$ , the compression of the buffer is  $x$ .

- (a) In a simple model of the motion, the only force affecting the truck during this motion is the thrust from the buffer.

- (i) Show that, while the truck is in contact with the buffer, the truck performs simple harmonic motion.

(2)

- (ii) Find, in terms of  $n$ , the period of this motion.

(1)

- (b) In a more realistic model, the motion of the truck is affected by a resistance force of magnitude  $mkv$ , where  $v$  is the speed of the truck and  $k$  is a positive constant.

- (i) Show that, while the buffer is being compressed,  $x$  satisfies the equation

$$\ddot{x} + K\dot{x} + n^2x = 0$$

(3)

- (ii) At time  $t = 0$ , the truck is travelling with speed  $U$ . Given that  $k = \frac{5n}{2}$  find  $x$  in terms of  $n$ ,  $U$  and  $t$ . (6)

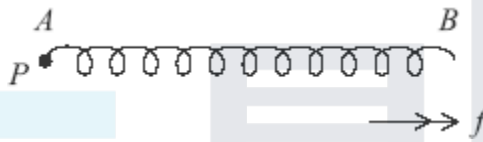
- (iii) By means of a sketch, or otherwise, explain whether the type of damping is light, critical or heavy. (2)

(Total 14 marks)

### Q3.

A light spring  $AB$  lies at rest and unstretched on a smooth horizontal surface. A particle,  $P$ , of mass  $m$  is attached to the end  $A$  of the spring.

The spring has stiffness  $2mn^2$ , where  $n$  is a positive constant. The end  $B$  of the spring is set in motion and moves with constant acceleration of magnitude  $f$  in the direction  $AB$ , as shown in the diagram. The particle  $P$  is consequently forced into motion.



At time  $t$  after the motion begins,  $P$  is moving with speed  $v$  and experiences a resistant force of magnitude  $3mnv$ . The extension of the spring is  $x$  and the displacement of  $P$  from its initial position in the direction  $AB$  is  $y$ .

- (a) Show that

$$y = \frac{1}{2}ft^2 - x$$

(3)

- (b) Hence show that  $P$  has velocity

$$ft - \frac{dx}{dt}$$

(1)

- (c) Hence show that

$$\frac{d^2x}{dt^2} + 3n\frac{dx}{dt} + 2n^2x = f + 3nft$$

(5)

- (d) Given that  $n = 1$ , and that a particular integral for this differential equation is

$$x = \frac{f}{4}(6t - 7)$$

find an expression for  $x$ , in terms of  $f$  and  $t$ .

**Q4.**

A particle  $P$ , of mass  $m$ , is suspended from a fixed point  $O$  by a light elastic string, of natural length  $a$  and modulus of elasticity  $5mn^2a$ , where  $n$  is a positive constant. When the particle hangs in equilibrium vertically below  $O$ , the extension of the string is  $\frac{g}{5n^2}$ . At time  $t = 0$ , the particle is projected vertically downwards from this equilibrium position with speed  $U$ .

During its subsequent motion, when  $P$  is moving with speed  $v$ , it experiences a resistance force of magnitude  $2mnv$ . The displacement of  $P$  below its equilibrium position at time  $t$  is  $x$ .

- (a) Show that, whilst the string is taut,  $x$  satisfies the equation

$$\frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 5n^2x = 0$$

(5)

- (b) Find  $x$  in terms of  $U$  and  $t$ .

(7)

- (c) State whether the nature of the damping caused by the resistance force is light, critical or heavy.

(1)

- (d) Find, in terms of  $U$  and  $\pi$ , the speed of  $P$  when it first returns to its equilibrium position.

(4)

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(Total 17 marks)

**Q5.**

A particle, of mass 2 kg, is suspended from a fixed point  $O$  by a light spring of natural length 0.5 metres and modulus of elasticity 49 N.

- (a) Initially, the particle hangs at rest in equilibrium below  $O$ . Find the extension of the spring in this position.

(2)

- (b) A force,  $F$  newtons, is then applied to the particle in a vertically downwards direction. The displacement of the particle below its equilibrium position at time  $t$  seconds later is  $x$  metres. Given that  $F = 12 \cos nt$ , where  $n$  is a positive constant, show that

$$\frac{d^2x}{dt^2} + 49x = 6 \cos nt$$

(5)

- (c) In the case where  $n = 5$ , find an expression for  $x$  at time  $t$ .

(10)

- (d) State the value of  $n$  for which resonance occurs.

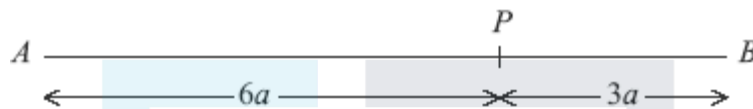
(1)

(Total 18 marks)

### Q6.

The ends of a light, uniform elastic string are fixed to two points,  $A$  and  $B$ , a distance  $9a$  apart on a smooth, horizontal plane. The string is of natural length  $6a$  and modulus of elasticity  $4mn^2a$ , where  $n$  is a constant.

A particle of mass  $m$  is attached to the string at  $P$ , where  $AP = 6a$ . The natural length of  $AP$  is  $4a$  and the natural length of  $BP$  is  $2a$ . In this position, the particle is in equilibrium.



The particle is moved a distance  $\frac{1}{2}a$  towards  $B$  and then released from rest at time  $t = 0$ . The displacement of the particle from its equilibrium position at time  $t$  is  $x$ .

Hence initially  $x = +\frac{1}{2}a$ .

The motion of the particle is resisted by a force of magnitude  $2mnv$ , where  $v$  is the speed of the particle at time  $t$ .

- (a) Show that  $x$  satisfies

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 3n^2x = 0$$

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(7)

- (b) Given that  $n = 1$ , find  $x$  in terms of  $a$  and  $t$ .

(8)

(Total 15 marks)

### Q7.

A particle  $P$ , of mass  $2m$ , is suspended from a fixed point  $O$  by a light elastic string of natural length  $a$  and modulus of elasticity  $8mn^2a$ , where  $n$  is a positive constant. The particle is released from rest when  $t = 0$  at a point  $A$ , where  $A$  is vertically below  $O$  and  $OA = a$ .

When the particle is moving with speed  $v$ , it experiences air resistance of magnitude  $4mnv$ .

- (a) The displacement of  $P$  below  $A$  at time  $t$  is  $x$ . Show that  $x$  satisfies the equation

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 4n^2x = g$$

(3)

- (b) Find  $x$  in terms of  $n$ ,  $g$  and  $t$ .

(12)

- (c) Hence show that the first time the particle comes to rest is when  $t = \frac{\pi}{\sqrt{3}n}$ .

(4)

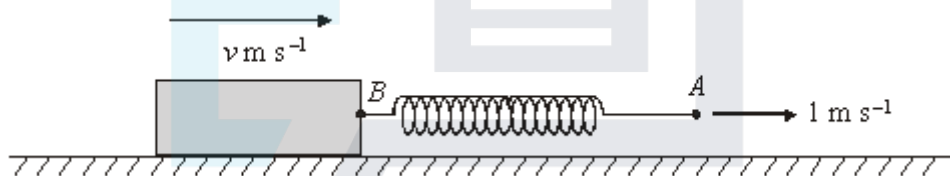
(Total 19 marks)

**Q8.**

A block of mass 0.5 kg moves on a smooth horizontal surface. It is attached to the end  $B$  of a light elastic spring  $AB$ , where  $AB$  is horizontal.

The natural length of the spring is 0.5 m and its stiffness is 5 N m<sup>-1</sup>.

The end  $A$  of the spring is pulled horizontally with constant speed of 1 m s<sup>-1</sup>, causing the block to move across the horizontal surface.



At time  $t$  seconds, the length of the spring is  $x$  metres and the block is moving with speed  $v$  m s<sup>-1</sup>.

The motion of the block is subject to a resistive force of  $v$  N.

- (a) Write down an expression for the tension in the spring in terms of  $x$ .

(2)

- (b) Show that the equation of motion of the block is

$$\frac{dv}{dt} + 2v - 10x = -5.$$

(3)

- (c) (i) Explain why  $\dot{x} = 1 - v$ .

(1)

- (ii) Hence show that

$$\ddot{x} + 2\dot{x} + 10x = 7.$$

(2)

- (d) Find the general solution of this equation.

(6)

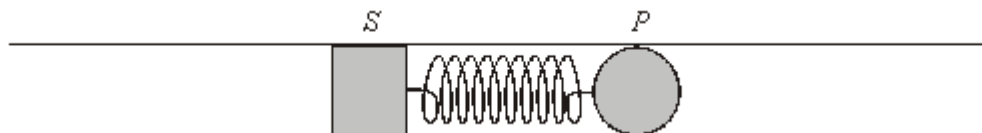
- (e) State the limiting value of  $x$ . Give a reason for your answer.

(2)

(Total 16 marks)

### Q9.

A solid ball  $P$  is attached to one end of a light elastic spring. The other end of the spring is attached to a piston  $S$  which is free to move horizontally inside a smooth tube, as shown in the diagram.



The mass of  $P$  is 0.5 kg. The spring has natural length 0.5 m and stiffness  $4.5 \text{ N m}^{-1}$ . The system initially rests in equilibrium.

The piston  $S$  is forced to oscillate, causing  $P$  to move. At time  $t$  seconds, the

displacements of  $S$  and  $P$  are  $\frac{1}{2} \sin 2t$  metres and  $x$  metres respectively, measured in the same direction from their initial positions.

- (a) Draw a diagram showing the forces acting on  $P$  and the displacements of  $P$  and  $S$ .

(1)

- (b) Show that  $x$  satisfies the differential equation

$$\ddot{x} + 9x = \frac{9}{2} \sin 2t.$$

(4)

- (c) Find  $x$  as a function of  $t$ .

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(9)

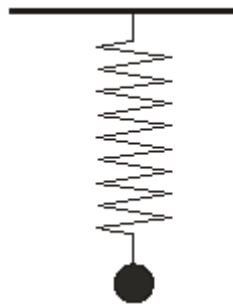
- (d) Show that  $P$  is instantaneously at rest when  $t = \frac{2\pi}{5}$ .

(2)

(Total 16 marks)

### Q10.

A sphere of mass 2 kg is fixed to one end of a spring. The other end of the spring is attached to the fixed point  $P$ , as shown in the diagram. The natural length of the spring is 0.5 metres.



- (a) When the system is in equilibrium, the length of the spring is 0.7 metres.

Find the modulus of elasticity of the spring.

(3)

- (b) The sphere is pulled down below its equilibrium position and released from rest. At time  $t$  seconds, the displacement of the sphere from its equilibrium position is  $x$  metres.

- (i) Show that

$$\frac{d^2 x}{dt^2} = -49x$$

(5)

- (ii) Find the value of  $t$  when the sphere returns to its equilibrium position for the first time.

(4)

(Total 12 marks)

### Q11.

An elastic string has natural length  $l$  and modulus of elasticity  $\lambda$ . One end of the string is fixed and a particle of mass  $m$  is attached to the other end of the string.

- (a) When the particle hangs vertically in equilibrium, the extension of the string is  $\frac{l}{4}$ . Show that  $\lambda = 4mg$ .

(2)

- (b) The particle is then set into vertical motion and  $x$  is the displacement of the particle from the equilibrium position at time  $t$ . Assume that the string remains taut during the vertical motion.

- (i) Show that  $\frac{d^2 x}{dt^2} = \frac{4gx}{l}$ .

(5)

- (ii) Hence find the period of the motion.

(2)

(Total 9 marks)

**Q12.**

A particle  $P$ , of mass  $m$ , is suspended from a fixed point  $O$  by a light elastic string of natural length  $2a$  and modulus  $4mn^2a$ , where  $n$  is a positive constant. The particle is released from rest at a point  $C$ , where  $C$  is vertically below  $O$  and  $OC = 2a$ .

When the particle is moving with velocity  $v$ , it experiences air resistance of magnitude  $2mnv$ .

- (a) The displacement of  $P$  below  $C$  at time  $t$  is  $x$ .

Show that  $x$  satisfies the equation

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 2n^2x = g$$

(4)

- (b) Find  $x$  in terms of  $n$ ,  $g$  and  $t$ .

(9)

(Total 13 marks)

**Q13.**

A spring, of natural length  $4a$  and modulus  $\lambda$ , has one end attached to a fixed support  $A$ , and a particle  $P$  of mass  $m$  is attached to its other end. Another spring, of natural length  $2a$  and modulus  $4mg$ , has one end attached to  $P$  and the other end attached to a fixed support  $B$ , which is situated at a distance of  $10a$  vertically below  $A$ . The system is in equilibrium in a vertical line with the upper spring stretched to a length of  $7a$  and the lower spring stretched to a length of  $3a$  as shown in the diagram.



- (a) Show that  $\lambda = 4mg$ .

(4)

- (b) At time  $t = 0$ , the particle is lowered to a distance  $\frac{a}{2}$  below its equilibrium position and released from rest. The subsequent motion of  $P$  is subject to a resistance of magnitude  $\frac{1}{5}mkv$ , where  $k^2 = \frac{6g}{a}$  and  $v$  is the speed of the particle at time  $t$ .

- (i) Given that  $x$  is the downward displacement of  $P$  from its equilibrium position at time  $t$ , show that



$$10 \frac{d^2 x}{dt^2} + 2k \frac{dx}{dt} + 5k^2 x = 0$$

(6)

- (ii) Hence find  $x$  in terms of  $a$ ,  $k$  and  $t$ .

(9)

(Total 19 marks)

**Q14.**

A particle  $P$  of mass  $m$  is suspended from a fixed point  $O$  by a light elastic string of natural length  $a$  and modulus  $6mn^2a$ , where  $n$  is a positive constant. The particle is released from rest at a point  $A$ , where  $A$  is vertically below  $O$  and distance  $OA$  is  $a$ .

When the particle is moving with velocity  $v \text{ m s}^{-1}$  it experiences air resistance of magnitude  $2mnv \text{ N}$ .

- (a) The displacement of  $P$  below  $A$  at time  $t$  is  $x$ .

Show that  $x$  satisfies the equation

$$\frac{d^2 x}{dt^2} + 2n \frac{dx}{dt} + 6n^2 x = g$$

(3)

- (b) Find  $x$  in terms of  $n$ ,  $g$  and  $t$ .

(10)

- (c) Hence show that the first time the particle comes to rest is when  $t = \frac{\pi}{\sqrt{5n}}$ .

(4)

(Total 17 marks)

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